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Lymphatic filariasis in Ghana: entomological investigation of transmission dynamics and intensity in communities served by irrigation systems in the Upper East Region of Ghana

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Summary We conducted an entomological study to document the effect of irrigation on the vectors and transmission dynamics of lymphatic filariasis in the Upper East Region of Ghana. Mosquitoes were collected by indoor spraying of houses in a cluster of communities located around irrigation projects (Tono and Vea) and others without reservoirs (Azoka). *Anopheles gambiae s.s.* was the dominant species and major vector, followed by *An. funestus*. *Anopheles arabiensis* constituted 9–14% of the *An. gambiae* complex but none were infective. *Culex quinquefasciatus* was also not infective in these communities. Chromosomal examinations showed that >60% ($n = 280$–386) of the *An. gambiae s.s.* in irrigated communities were Mopti forms whilst 73% ($n = 224$) in the non-irrigated area were Savannah forms. Infectivity rates (2.3–2.8 *vs.* 0.25), worm load (1.62–2.04 *vs.* 1.0), annual bites per person (6.50–8.83 *vs.* 0.47) and annual transmission potential (13.26–14.30 *vs.* 0.47) were significantly higher in irrigated communities.

keywords *Wuchereria bancrofti,* vectors, transmission, irrigation, Ghana

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Introduction

Lymphatic filariasis, caused by infection with the mosquito-borne filarial nematode *Wuchereria bancrofti,* is a deforming parasitic disease that affects over 100 million people in more than 70 tropical and subtropical countries (Ottesen & Ramachandran 1995). In countries where lymphatic filariasis is well established, the prevalence of infection continues to increase, primarily because of unplanned growth of cities and water resource development such as irrigation, which creates numerous breeding sites for the mosquitoes that transmit the disease (Service 1984).

Surveys in Ghana have indicated that Bancroftian filariasis is present in most parts of the country, with considerable regional variations in prevalence. Studies along the coast of Ghana have shown endemic foci along the west coast of the country with overall microfilaria (mf) prevalences of 9–25% and microfilarial intensities of 321–1172 mf/ml of blood. Hydrocele affected 8.5–27%, whilst elephantiasis of the limbs affected 5.6–6.6% of the population (Dunyo *et al.* 1996). Hunter (1992) observed that a number of communities in Ghana with filariasis had water impoundment, such as small dams provided for agricultural purposes; in Okyereko, an irrigation project community in the Central Region of Ghana, the prevalence of microfilaraemia was 26.4%; 13.8% had hydrocele and 1.4% had elephantiasis (Dzodzomenyo *et al.* 1999). *Anopheles gambiae s.l.* and *An. funestus* were identified as the vectors. Gyapong *et al.* (1994) also found a high prevalence of clinical manifestations (32% hydrocele, 3.6% limb elephantiasis) and a microfilaraemia prevalence of 32.4% in people living close to a large irrigation scheme (Tono) in the Upper East Region of Ghana. However, there were no entomological investigations to identify the role irrigation plays in the transmission dynamics and intensity

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of the disease in the region, and we conducted this study to identify the vectors of Bancroftian filariasis and document the transmission characteristics of the disease with respect to the irrigation projects in the Upper East Region of Ghana.

Study area

The study took place in three districts, Kassena Nankana (KND), Bolgatanga (BGD) and Bawku (BKD) in the Upper East Region of Ghana, near the border with Burkina Faso. The vegetation in these areas is dry Guinea savannah characterized by short grass and some fire-resistant trees. The climate is sub-Sahelian, with a mean minimum and maximum temperature of 20 and 40 °C, respectively. The annual rainfall is 850–1000 mm in the months of May–September, followed by a prolonged dry season. There are two water reservoirs, a large one at Tono in KND and a medium-sized one at Vea in BGD. Each of these projects serves eight villages with a total farmer population of 6000. The maximum surface area of Tono reservoir is 1860 ha with a maximum storage of 93×10^6 m^3 serving 32 km of main canals; Vea has a maximum surface area of 405 ha with a maximum storage of 17×10^6 m^3 serving 21 km of main canals. The irrigation projects were constructed to provide water for livestock and to facilitate dry season farming. The study was conducted in Korania, Saboro and Wuru near the Tono irrigation project in KND; and in Vea and Kongo near the Vea reservoir in BGD. Another community, Azoka, and other nearby settlements located in BKD, about 70 km east of the two irrigation projects, were also included. The study areas are mainly rural with settlements surrounded by farmland. Most people live in multifamily compounds.

Materials and methods

Indoor resting mosquitoes were collected at the beginning of the dry season to provide an indirect estimate of biting densities. White sheets were spread on the floors of rooms before spraying with pyrethrum insecticide. After 10–15 min knocked-down mosquitoes were picked from the sheets and sorted by species based on their morphological characteristics (Gillies & De Meillon 1968; Gillies & Coetzee 1987). Each mosquito was put on a slide and the head, thorax and abdomen were macerated separately in a drop of saline under a stereo microscope at 20× magnification and examined under a compound microscope at 100× magnification for filarial worms (WHO 1987). Third-stage (L_3) larvae of *W. bancrofti* were identified morphologically (Nelson 1959); those which were broken during dissection and therefore could not be identified were preserved according to the technique of Toe *et al.* (1998). *W. bancrofti* DNA was identified using PCR in a final reaction mix of 50 µl with two oligonucleotide primers, NV-1 and NV-2 (Ramzy *et al.* 1997). The PCR products were electrophoresed on 2% agarose gel after staining with ethidium bromide. Microfilariae from microfilaraemic individuals were used as positive controls, with *Plasmodium falciparum* DNA used as negative controls. The ovaries of half-gravid mosquitoes were dissected and processed for polytene chromosomes in accordance with the technique of Hunt (1973) and the nomenclature based on Coluzzi *et al.* (1979).

Entomological parameters were calculated as follows:

- Infection rate = proportion of mosquitoes carrying *W. bancrofti* larvae of any stage (L_1–L_3).
- Infectivity rate = proportion of mosquitoes containing at least one infective (third stage) larva.
- Annual biting rate (ABR) = number of blood-fed mosquitoes caught in rooms × 365/(total number of people sleeping in rooms × number of captures) (endophily is assumed to be 100%).
- Annual infective biting rate (AIBR) = ABR × infectivity rate.
- Worm load = total number of L_3/total number of mosquitoes carrying L_3.
- Annual transmission potential (ATP) = AIBR × worm load.

Results

Table 1 shows the abundance of mosquitoes caught in the cluster of villages around the two irrigated facilities and the non-irrigated area in the Upper East Region of Ghana. About 4900 mosquitoes (1534 from Tono, 2504 from Vea, 862 from Azoka) were collected, comprising five species of *Anopheles*, one species of *Culex* and one species of *Aedes*. Of these, *An. gambiae s.l.* constituted more than 70% of all collections in the study areas, followed by *An. funestus* and *Cx. quinquefasciatus*. *An. gambiae s.s.* and *An. arabiensis* were identified as the two sibling species of the *An. gambiae* complex occurring sympatrically, with *An. arabiensis* dominating in the study areas (Table 2). The proportions of the two members of the *An. gambiae* complex in all villages were not significantly different ($P > 0.05$). The analysis of chromosomal polymorphism of *An. gambiae s.s.* in the study areas indicates that the Savanna and Mopti forms coexist (Table 3). They are characterized by different inversion frequencies, which exist in variable proportions. The savanna population was characterized by higher frequencies of inverted arrangements of 2Rb and 2La chromosomes (Coluzzi *et al.* 1985).

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Table 1 Mosquitoes collected from irrigated communities (Tono and Vea) and Azoka, including nearby settlements without reservoirs in the Upper East Region of Ghana

	Tono		Vea		Azoka	
Species	No.	%	No.	%	No.	%
An. gambiae s.l.	1256	81.9	1831	73.1	756	87.7
An. funestus	254	16.6	471	18.8	48	5.6
An. pharoensis	0	0	27	1.1	2	0.23
An. nili	0	0	14	0.6	0	0
An. rufipes	24	1.6	0	0	0	0
Cx. quinquefasciatus	0	0	128	5.1	51	5.9
Ae. aegypti	0	0	33	1.3	5	0.6
Total	1534		2504		862	

Table 2 Distribution of sibling species of *Anopheles gambiae* complex in irrigated communities (Tono and Vea) and Azoka, including nearby settlements without water reservoirs in the Upper East Region of Ghana

Locality	No. examined	No. of *An. gambiae s.s.* (%)	No. of *An. arabiensis* (%)
Tono	464	402 (86.6)	62 (13.4)
Vea	618	562 (90.9)	56 (9.06)
Azoka	356	306 (85.9)	50 (14.0)
Total	1438	1270 (88.3)	168 (11.7)

Table 3 Frequencies of Mopti and Savanna chromosomal population forms of *Anopheles gambiae s.s.* in irrigated communities (Tono and Vea) and Azoka, including nearby settlements without reservoirs in the Upper East Region of Ghana

Locality	No. examined	No. of Mopti (%)	No. of Savanna (%)
Tono	464	280 (69.7)	122 (30.4)
Vea	618	386 (62.5)	176 (37.5)
Azoka	356	82 (26.8)	224 (73.2)
Total	1438	748	320

The Mopti population was characterized by the polymorphism of 2Rbc/2Ru and by almost complete fixation of the inverted arrangement 2 La (Touré 1989). The proportions of Mopti forms were greater in irrigated communities, whilst the reverse was the case in Azoka, the community without a reservoir (Table 3).

Table 4 shows the infectivity rates of the two most prevalent mosquito species in the study areas. *Anopheles gambiae s.l.* recorded infectivity rates of 3.34, 2.51 and 0.26% in the Tono, Vea and Azoka, respectively, followed by 1.27% infectivity rate for *An. funestus* in Vea. Infectivity rates of both species were higher in the irrigated areas ($P < 0.01$). Of 90 (42 from Tono, 46 from Vea and 2 from Azoka) *An. gambiae s.l.* which were infective with L_3, only 7 and 11 which were half-gravid could be differentiated into Mopti and Savanna forms, respectively. Although these numbers are small, we can exclude the hypothesis that only one of the forms can be a vector. None of the 168 *An. arabiensis* and 179 *Cx. quinquefasciatus* encountered in these areas were infective. PCR confirmed (Figure 1) 94.6% (71/75) of the infective larvae dissected as *W. bancrofti*. The DNA of four of the larvae could not be amplified.

Worm loads [the average number of infective larvae (L_3) per infective mosquito] were 1.62, 2.04 and 1.0 for *An. gambiae s.l.* in Tono, Vea and Azoka, respectively, and 2.0 for *An. funestus* in Vea (Table 4). The estimated annual numbers of *An. gambiae s.l.* and *An. funestus* bites per person (ABR) were 315 in Tono, 283 in Vea and 188 in Azoka. The potential number of infective bites per person per year (AIBR) in the irrigated communities, 8.83 in Tono and 6.50 in Vea, were significantly higher ($P < 0.001$) than the 0.47 estimated for Azoka. The annual transmission potential (ATP) which was 14.30 in Tono and 13.26 in Vea, was also significantly ($P < 0.001$) higher than the 0.47 estimated in Azoka (Table 4).

Discussion

This study identified *An. gambiae s.s.* and *An. funestus* as the dominant species, with *An. gambiae s.s.* being the major vector of Bancroftian filariasis in the study areas. These findings are similar to those recorded in rural parts of coastal Ghana (Dunyo *et al.* 1996) where *An. gambiae s.s. and An. melas* were the two sibling species of the *An. gambiae* complex that were infective. In our study, *An. arabiensis* coexisted with *An.gambiae s.s.* but no infective specimen of *An. arabiensis* was found, implying that it is not a vector in this area, but this could have been an artefact because only indoor resting mosquitoes were sampled, which could underestimate the numbers of this species less probable to rest indoors than outdoors (Bryan *et al.* 1987). We used spray sheet collection in this study to

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Table 4 Entomological parameters for the transmission of Bancroftian filariasis in irrigated communities (Tono and Vea) and Azoka, including nearby settlements without irrigation in the Upper East Region of Ghana

Locality	Species	No. dissected	No. with L_3	Infective (%)	Total no. of L_3	ABR	AIBR	Worm load	ATP
Tono	*An. gambiae s.l.*	1256	42	3.34	68				
	An. funestus	254	0	0	0				
	Total	1510	42	2.80	68	315	8.83	1.62	14.30
Vea	*An. gambiae s.l.*	1831	46	2.51	94				
	An. funestus	471	6	1.27	12				
	Total	2302	52	2.30	106	283	6.50	2.04	13.26
Azoka	*An. gambiae s.l.*	756	2	0.26	2				
	An. funestus	48	0	0	0				
	Total	804	2	0.25	2	188	0.47	1.0	0.47

L_3: infective (third stage) larva of *Wuchereria bancrofti*; ABR: annual biting rate; AIBR: annual infective biting rate; ATP: annual transmission potential.

obtain large numbers of mosquitoes, because this method collects resting blood-fed and gravid female mosquitoes in addition to hungry unfed females in search of blood meal, which are the majority of those obtained from human landing catches (Mwandawiro *et al.* 1997). However, it is expected that fewer mosquitoes from spray catches would be infective than those from human landing catches as infective larvae are lost during feeding (McMahon *et al.* 1981). Biting rates were also estimated from indoor catches because the vectors in the study areas are highly endophilic (F. Binka, personal communication). The higher proportion of Mopti forms of *An. gambiae s.s.* occurring in the communities served by irrigation is not surprising because these chromosomal population forms have remarkable ecological flexibility and are known to prevail in inundated areas where dry season breeding opportunities exist (Coluzzi *et al.* 1985; Petrarca 1987; Appawu *et al.* 1994).

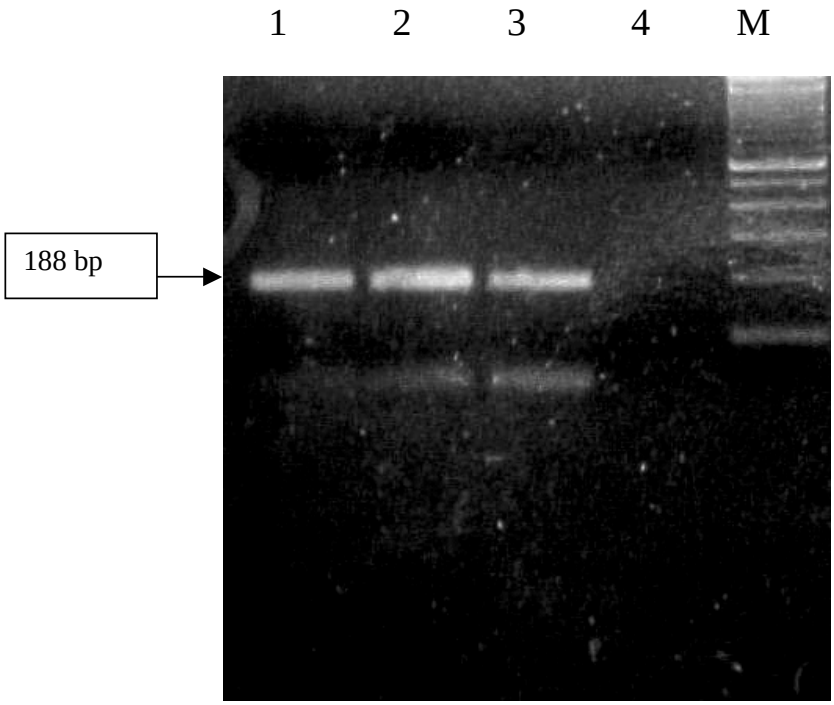


Figure 1 Detection of *Wuchereria bancrofti* DNA using PCR assay. The amplified 188 base pair (bp) product was electrophoresed in a 2% agarose gel and visualized by staining with ethidium bromide. Lane 1, positive control; Lane 2–3, positive samples; Lane 4, negative control; Lane 5, M, 100 bp 'ladder' (molecular size marker).

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Culex quinquefasciatus, which is a vector of lymphatic filariasis in East Africa (White 1971; Wijers & Kiilu 1977; Pedersen *et al*. 1999) was not infective in this study and therefore may not have any vectorial role in irrigated or dry communities. This observation is consistent with previous ones made in coastal Ghana (Dunyo *et al*. 1996; Dzodzomenyo *et al*. 1999). The susceptibility of *Cx. quinquefasciatus* from Ghana to local strains of *W. bancrofti* infection has not been tested; but *Cx. quinquefasciatus* from Liberia were found not to be susceptible to local strains of *W. bancrofti* (Jayasekera *et al*. 1980).

This study has shown that there is transmission of Bancroftian filariasis in Kassena Nankana, Bolgatanga and Bawku. Both annual infective bites and annual transmission potential were higher in the irrigated communities. Microfilaria prevalence rates of 23.4% were recorded in the Tono area (Gyapong *et al*. 1994), and a preliminary survey with ICT cards estimated infection rates of 12.5% for Vea and 0% for Azoka (unpublished report). Relatively high microfilarial rates in the human population coupled with somewhat higher vector densities in irrigated communities led to more infective feeds for mosquitoes and hence a high intensity of disease transmission. In Okyereko, another irrigated community in southern Ghana, the opening of the canals for farming activities could cause the dry season populations of vectors of lymphatic filariasis to increase to the levels of the wet season (Dzodzomenyo *et al*. 1999). Hence vector density is probable to differ more between irrigated and non-irrigated areas later in the dry season than we observed in our collections early in the dry season.

The irrigation projects provide not only the means of growing crops throughout the year in a harsh arid environment, but also create large expanses of perennial water: ideal breeding sites for mosquitoes with higher humidity which probably favours vector survival for an age to which they can become infective. Irrigated areas attract people, leading to overcrowding and slum settlements. Clearly irrigation will become increasingly important in food production for an ever-expanding human population. However, to augment the current plan of WHO to eliminate filariasis by mass treatment of the human population with ivermectin/DEC and albendazole, water resource development agencies and health policy makers should collaborate in the planning and execution of irrigation schemes in order to reduce vector breeding.

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References

Appawu MA, Baffoe-Wilmot A, Afari EA, Nkrumah FK & Petrarca V (1994) Species composition and inversion polymorphism of the *Anopheles gambiae* complex in some sites of Ghana, West Africa. *Acta Tropica* **56**, 15–23.

Bryan JH, Petrarca V, Di Deco M & Coluzzi M (1987) Adult behaviour of members of *Anopheles gambiae* complex in The Gambia with special reference to *Anopheles melas* and its chromosomal variants. *Parassitologia* **29**, 221–249.

Coluzzi M, Sabatini A, Petrarca V & Di Deco MA (1979) Chromosomal differentiation and adaptation to human environments in the *Anopheles gambiae* complex. *Transactions of the Royal Society of Tropical Medicine and Hygiene* **73**, 483–493.

Coluzzi M, Petrarca V & Di Deco MA (1985) Chromosomal inversion intergradation and incipient speciation in *Anopheles gambiae*. *Bolletino Di Zoologia* **52**, 45–63.

Dunyo SK, Appawu M, Nkrumah FK, Baffoe-Wilmot A, Pedersen EM & Simonsen PE (1996) Lymphatic filariasis on the coast of Ghana. *Transactions of the Royal Society of Tropical Medicine and Hygiene* **90**, 634–638.

Dzodzomenyo M, Dunyo SK, Ahorlu CK *et al*. (1999) Bancroftian filariasis in an irrigation project community in southern Ghana. *Tropical Medicine and International Health* **4**, 13–18.

Gillies MT & Coetzee M (1987) *A Supplement to the Anophelinae of Africa South of the Sahara*. South African Institute for Medical Research, Johannesburg.

Gillies MT & De Meillon B (1968) *The Anophelinae of Africa South of Sahara (Ethiopian Zoogeographical Region)*, 2nd edn. South African Institute for Medical Research, Johannesburg.

Gyapong JO, Magnussen P & Binka FN (1994) Parasitological and clinical aspects of Bancroftian filariasis in the Kassena-Nankana District, Upper East Region, Ghana. *Transactions of the Royal Society of Tropical Medicine and Hygiene* **88**, 555–557.

Hunt RH (1973) A cytological technique for the study of *Anopheles gambiae* complex. *Parassitologia* **15**, 137–139.

Hunter JM (1992) Elephantiasis: a disease of development in North East Ghana. *Social Science and Medicine* **35**, 627–649.

Jayasekera N, Curtis CF, Zielke E, Kuhlow F, Jansen CG & Chelliah RV (1980) The susceptibility of Liberian *Culex quinquifasciatus* to *Wuchereria bancrofti* in Sri Lanka. *Tropenmedizin und Parasitologie* **4**, 507–511.

McMahon JE, Magayuka SA, Kolstrup N, Mosha FW, Bushford FM & Abaru DE (1981) Studies on the transmission and prevalence of Bancroftian filariasis in four coastal villages of Tanzania. *Annals of Tropical Medicine and Parasitology* **75**, 415–431.

Mwandawiro CS, Fujimaki Y & Katsivo M (1997) Mosquito vectors of Bancroftian filariasis in Kwale District, Kenya. *East African Medical Journal* **74**, 288–293.

Nelson GS (1959) The identification of infective filarial larvae in mosquitoes: with a note on the species found in 'wild' mosquitoes on the Kenyan Coast. *Journal of Helminthology* **33**, 233–257.

Ottesen EA & Ramachandran CP (1995) Lymphatic filariasis infection and disease: control strategies. *Parasitology Today* **11**, 129–131.

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Pedersen EM, Kilama WL, Swai ABM, Kihamia CM, Rwiza H & Kisumku UM (1999) Bancroftian filariasis on Pemba Island, Zanzibar, Tanzania: an update on the status in urban and semi-urban communities. *Tropical Medicine and International Health* **4**, 295–301.

Petrarca V, Vercruysse J & Coluzzi M (1987) Observations on the *Anopheles gambiae* complex in Senegal River Basin, West Africa. *Medical and Veterinary Entomology* **1**, 303–312.

Ramzy RM, Farid HA, Kamal IH *et al.* (1997) A polymerase chain reaction-based assay for detection of *Wuchereria bancrofti* in human blood and *Culex pipiens*. *Transactions of the Royal Society of Tropical Medicine and Hygiene* **91**, 156–160.

Service MW (1984) Problems of vector-borne disease and irrigation projects. *Insect Science and its Application* **5**, 227–231.

Toe L, Adjami A, Bissan Y & Houggard JM (1998) A case of cross infection by *Onchocerca volvulus* and *Onchocerca orchengi* in *Simulium damnosum* s.l. *Medicine Tropicale* **58**, 269–270.

Touré YT (1989) Vector study and vector control in the control of malaria and babesiosis: the current state of studies of malaria vectors and anti-vectorial campaign in West Africa. *Transactions of the Royal Society of Tropical Medicine and Hygiene Supplement* **83**, 39–41.

White GB (1971) Studies on transmission of bancroftian filariasis in North-Eastern Tanzania. *Transactions of the Royal Society of Tropical Medicine and Hygiene* **65**, 819–829.

WHO (1987) *Control of Lymphatic Filariasis. A Manual for Health Personnel*. World Health Organization, Geneva.

Wijers DJB & Kiilu G (1977) Bancroftian filariasis in Kenya III. Entomological investigations in Mambrui, a small coastal town, and Jaribuni, a rural area more inland (Coastal Province). *Annals of Tropical Medicine and Parasitology* **71**, 347–359.

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